

INSTRUCTION MANUAL

Kill Dozer



WARNING:

CHOKING HAZARD – Small Parts.

Key Features

- Motor – driven
- Drive within black circular area limited by white line
- Push other robots out of area
- Ultrasonic and ToF distance sensor to detect other robots
- TCS34725 RGB colour sensor to not exit white line area
- Hardware modifiable by user

IMPORTANT: Read Instructions carefully

Note: The manual is intended for users with a basic understanding of electrical and mechanical engineering. Failure to comply with the manual may cause damage to the device or user.

Package Contents

- 1 x User manual
- 1 x Sumo Bot
- 1 x Lithium Polymer (LiPo) battery
- 1 x ESP32 TTGO microcontroller

Specifications

- Robot dimensions (L x W x H) (cm): 25 x 20 x 25
- Weight (kg): 1
- Maximum Speed (m/s): 10
- Battery capacity: 11.1V 2600mAh

SAFETY PRECUATIONS

- **DO NOT** pick up the robot while it is moving.
- Keep hands, hair, and loose clothing away from the moving parts when the power is on.
- Always use fully charged batteries.
- **DO NOT** operate the robot on streets. Avoid contact with people, animals, or household objects.
- **DO NOT** enter the operating area of the robot while it is switched on.
- **DO NOT** submerge the robot into water as this may damage its electrical components.
- Make sure all sensors are clean and unobstructed.
- Avoid placing the robot on uneven or unstable surfaces.

CONTROL SPECIFICATIONS

MOTOR SPEED

The Speed is dynamically adjusted based on the distance measured by the ultrasonic sensor. Distances between 10 and 100cm are linearly mapped to an integer speed value between 30 and 255 using pulse width modulation (PWM).

If the objects are close by, the robot slows down.

If objects are further away the robot accelerates.

In competitive mode, when the opposing robot is at a speed of 50 or more, the robot overrides the mapped speed and sets its speed to maximum (255) to initiate charge.

STEERING

The ultrasonic sensor is used to trigger steering behavior. If an obstacle is 15cm or more from the robot it drives forward at a mapped speed.

TROUBLESHOOTING		
ISSUE	POSSIBLE CAUSES	SOLUTION
Robot does not switch on or powers off during operation.	-Battery is uncharged, damaged, or not properly connected. -Faulty internal power wiring.	-Fully charge or replace the battery. -Check internal power connections.
Robot does not respond to commands.	-Loose or damaged hardware connections. -Logic error or bug in software.	-Secure all hardware connections. -Review and debug software control logic.
Robot moves erratically or inconsistently.	-Battery is damaged or low. -Motor malfunction.	-Fully charge or replace the battery. -Inspect motors.
Robot fails to detect obstacles.	-Ultrasonic sensor is misaligned, obstructed. -Sensor interference from ambient noise, temperature, or lighting.	-Clean sensor surface. -Recalibrate or replace the sensor.

MAINTENANCE AND CARE

MAINTENANCE AND STORAGE REQUIREMENTS

- Clean the robot regularly using a dry microfiber towel.

Ensure all sensors are clean and unobstructed.

- Store robot indoors in a dry environment with temperatures between 10°C to 30°C.
- Avoid exposure to heat, moisture, or humidity.

BATTERY SPECIFICATIONS

CHARGING

Only use a Lithium Polymer (LiPo) – specific charger. If possible, remove the battery from the robot before charging. Never leave the battery unattended while charging to avoid overcharging as it might damage the battery or start a fire. For safe charging, use a 1C charging rate, which fully charges the battery in one hour. To maintain battery performance, use the

battery shortly after charging. If the battery will not be used for a while, discharge it to the recommended storage voltage which is 3.7V per cell for a LiPo 11.1V battery.

REPLACEMENT

A typically LiPo battery has a lifespan of approximately 300 charges, through this can vary depending on usage, charging habits and care. Immediately replace the battery if it becomes hot, swollen or visibly damaged. These are indicators that the battery is no longer safe to use. Batteries in this condition should be correctly disposed of according to e – waste and hazardous waste guidelines. Additionally, many LiPo chargers include an internal resistance check feature. If the internal resistance varies significantly between cells, this may indicate cell imbalance, and the battery should be replaced.

1. Power off the robot
2. Carefully disconnect the existing battery.
3. Dispose of the old battery following LiPo battery regulations.
4. Insert a new, fully charged battery ensuring correct polarity (match the + to + terminals and – to -).

WARRANTY

A warranty claim can be made on manufacturing faults within 12 months from the date of purchase. If the claim is approved, necessary replacement parts will be provided with no cost to the user.

The warranty does not cover the following:

- Damage resulting from wear and tear.
- Damage caused by Improper use, mishandling or failure to follow the user manual.
- Damage incurred during unauthorized maintenance or repair.
- Issues related to software.

How to submit a warranty claim:

1. Provide valid proof of purchase.
2. Describe the defect in detail, including photos or videos if possible.
3. Contact customer support.

OPERATING INSTRUCTIONS

1. Connect robot to a full charged LiPo battery or USB power supply.
2. Place the robot in a clear operating area, free from unintended obstacles.
3. Press the right button on the TTGO controller to start the system.
4. Wait for the start - up message to appear on the TTGO LCD screen before operating.